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ORIGINAL ARTICLE

Parenchymal FLAIR hyperintensity before thrombolysis is a prognostic factor of ischemic stroke outcome at 3 Tesla



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KEYWORDS

Acute stroke;
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Summary

Background: The goal of the present study was to determine whether the presence or absence of parenchymal FLAIR hyperintensity alone, before thrombolysis, might be a predictive factor of ischemic stroke outcomes after the acute phase of stroke and at 3 months.

Materials and methods: We retrospectively included 84 patients with an ischemic stroke between November 2007 and March 2012, who underwent 3 T MRI, were treated by thrombolysis, and had medical follow-up at 3 months. Two readers analyzed parenchymal FLAIR visibility. Logistic regressions were performed for NIHSS difference (NIHSS at admission – NIHSS at the end of hospitalization) and for 3 months modified Rankin Scale (mRS). Predictive values of positive parenchymal FLAIR for identifying poor outcome at discharge and at 3 months were estimated. **Results:** Parenchymal FLAIR positivity was not predictive of NIHSS difference but it predicted poor outcome at 3 months (sensitivity: 0.49 [0.37–0.60], specificity: 0.69 [0.46–0.91], positive predictive value: 0.87 [0.76–0.98] and negative predictive value: 0.24 [0.12–0.36]).

Conclusions: At 3 Tesla, the presence of a parenchymal hyperintense FLAIR signal before thrombolysis is predictive of a poor clinical outcome at 3 months.

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Abbreviations: FLAIR, Fluid attenuated inversion recovery; iv-tPA, Intravenous recombinant tissue plasminogen activator; MCA, Middle cerebral artery; DWI, Diffusion weighted imaging; NIHSS, National Institutes of Health Stroke Scale; mRS, Modified Rankin Scale; DWI-ASPECTS, Diffusion-weighted imaging – the Alberta Stroke Program Early Computed Tomography Score; Se, Sensitivity; Sp, Specificity; PPV, Positive predictive value; NPV, Negative predictive value.

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Introduction

Parenchymal FLAIR signal has proved to be helpful to determine if ischemic stroke is less than 4.5 hours or not, with potential usefulness in evaluating wake up stroke patients [1–3]. Indeed, International guidelines dissuade thrombolytic treatment of stroke with unknown onset time [4]. The maximal recommended delay in initiating thrombolysis is 4.5 hours after the onset of stroke [5,6]. The use of a common time window for initiating thrombolysis is based on the assumption that the cerebral parenchyma in all patients would support the same duration of hypoperfusion, which is yet to be proven. Furthermore, it seems that collateral blood flow could partly explain this inter individual variability by attenuating the effects of arterial occlusion [7]. Parenchymal FLAIR signal could be a better indicator as it determines tissue viability rather than hypoperfusion duration. Even though parenchymal FLAIR hyperintensities of ischemic lesions before thrombolysis are not predictive of treatment response (iv-tPA or mechanical thrombectomy) 1 day after treatment but parenchymal associated with vascular hyperintensities could predict poor ischemic stroke outcomes at 3 months [8–11]. Mechanical thrombectomy were made under conscious sedation [12]. Furthermore, some authors suggested that intravascular FLAIR hyperintensity is prognostic of good outcome at 3 months [7]. Thus, the association of vascular FLAIR hyperintensity (due to slow blood flow), and the parenchymal FLAIR hyperintensity, (due to vasogenic edema of the lesion), is not clear [13,14]. Therefore, it is important to analyze these two phenomena independently.

The goal of the present study was to determine whether the parenchymal FLAIR hyperintensity before thrombolysis might be a predictive factor of ischemic stroke outcomes after the acute phase of stroke (at hospital discharge) and at 3 months.

Materials and methods

Patients

We retrospectively evaluated data from patients who were treated for acute ischemic stroke in the MCA territory in our institution between November 2007 and March 2012. Patients were included if they had undergone 3 Tesla MRI with DWI and FLAIR sequences, and were treated, after their MRI, by intravenous fibrinolysis. All patients were listed in the stroke unit registry of our institution. The exclusion criteria were: unknown time of stroke onset, presence of major artifacts precluding FLAIR images analyses, and the absence of medical follow-up at 3 months for surviving patients. This retrospective data analysis had the approval of the local institutional review board of university hospitals of Reims. Data collected up to 2009 included patients treated with intravenous thrombolysis within 3 hours of symptom onset. Post 2009 data included patients treated up to 4.5 hours of symptom onset based on scientific evidence supporting extending the time window for thrombolysis [5]. According to the study design, written informed consent was waived but the patients or their next-of-kin gave informed consent for participation.

Patient information was anonymized and de-identified prior to analysis.

Patient data collected included demographics (gender and age), NIHSS at admission, systolic and diastolic blood pressure at admission, glycemia at admission, duration of hospitalization, and stroke symptom onset time to determine stroke-to-MRI and stroke-to-treatment delays. Stroke etiology was classified by subtype using the Trial of ORG 10172 in Acute Stroke Treatment (TOAST) criteria:

- large artery atherosclerosis;
- cardioembolism;
- small vessel occlusion;
- other determined etiology;
- undetermined etiology [15].

Neurologic functional outcomes were defined as the NIHSS at discharge and mRS at 3 months both of which were assessed by a neurologist [16]. The difference in NIHSS scores between admission and discharge were computed and categorized as:

- died patients;
- negative difference;
- 0 to 5;
- 6 to 10;
- 11 to 15;
- 16 to 20;
- 21 to 25.

mRS outcomes at 3 months were considered good if it was < 2 (patient with no significant disability), and poor if it was ≥ 2 (patient unable to carry out all previous activities).

MRI protocol

The MRI study was performed on a 3 T whole body clinical scanner with a 16-channel head coil. The protocol included the following 2 sequences: DWI sequence with a b-value of 0 and 1000 s/mm², repetition time = 2500 ms, echo time = 55 ms, field of view = 230 × 230 mm, matrix size = 256 × 256, 28 slices, and slice thickness = 5 mm without gap; and FLAIR sequence with repetition time = 11 000 ms, echo time = 125 ms, inversion time = 2800 ms, field of view = 230 × 230 mm, matrix size = 512 × 512, 28 slices, and slice thickness = 5 mm without gap.

Image analysis

MRI images were read by two radiologists (OT, FP), with more than 10 years experience in neuroradiology, to determine if the stroke lesions were FLAIR positive or not (Fig. 1). They were unblinded of the clinical information and the DWI images to better evaluate the lesion of interest. In case of multiple ischemic lesions with different FLAIR signal, patient was excluded (Fig. 2) to ensure that the clinical symptoms corresponded to the analyzed lesion and to be certain of the lesion age. The two readers first identified the DWI lesion followed by a review of the FLAIR images, focusing their analysis on the parenchymal FLAIR signal. Due

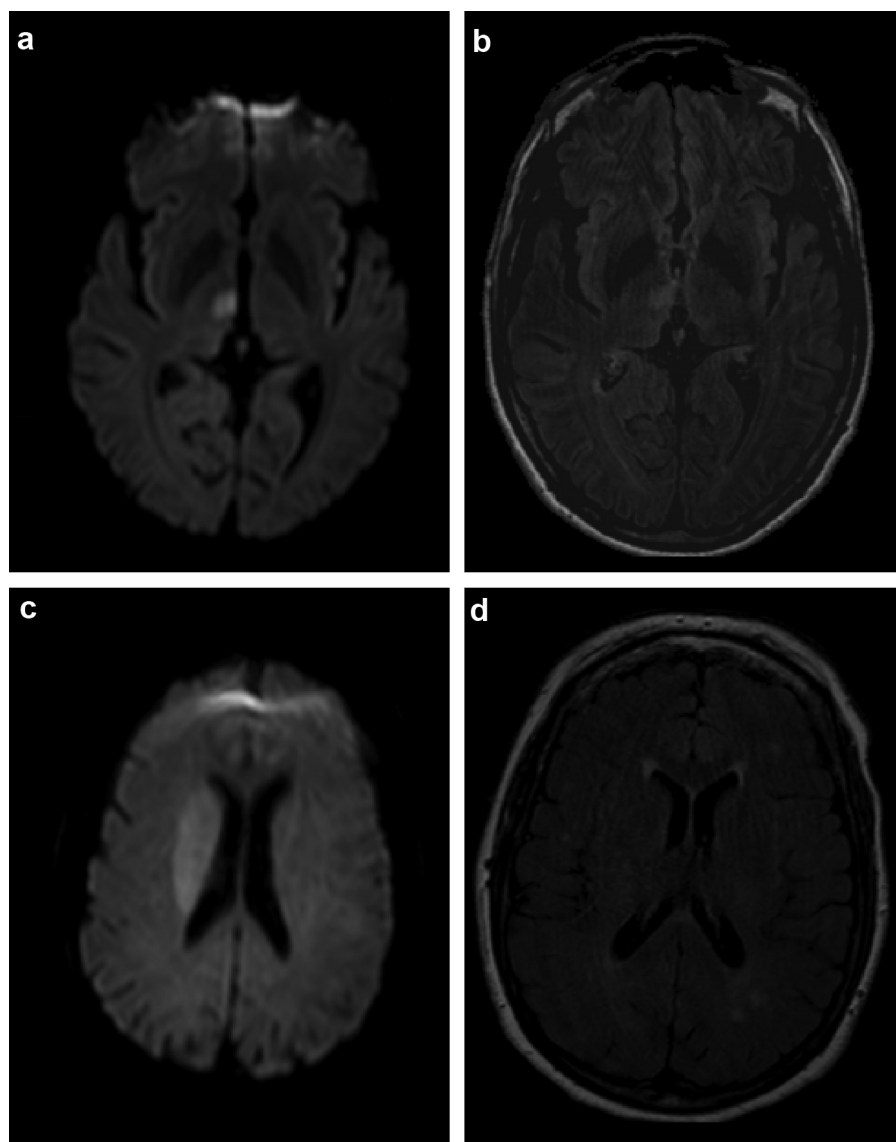


Figure 1 Patients were classified in 2 groups: positive FLAIR group (first line) corresponding to a parenchymal positive FLAIR signal (b) of ischemic lesion determined on DWI (a), and negative FLAIR (second line) corresponding to a parenchymal positive FLAIR signal (d) of ischemic lesion determined on DWI (c).

to the difficulty to assess parenchymal FLAIR intensity on some lesions, in case of doubt, positivity was determined by comparison to the contralateral side [17]. The readers were instructed not to consider intravascular FLAIR signal for the classification (Fig. 3). When the readers' assessments differed, a consensus was reached between the two readers. A third reader (SE) estimated the presence of leukoaraiosis on FLAIR images, according to the Fazekas scale: 0) no white matter lesions; 1) multiple punctuate lesions; 2) early confluence of lesions; and 3) large confluent lesions [18]. The same reader also estimated the DWI-ASPECTS Score of lesions seen on diffusion-weighted images [19,20]. Indeed, important leukoaraiosis and stroke volume can be confusing parameters to evaluate positivity of FLAIR signal.

Statistical methods

Age, gender, NIHSS at admission, stroke subtypes, stroke-to-thrombolysis delay, stroke-to-MRI delay, Fazekas' scale, ASPECTS score, systolic and diastolic blood pressure, glycemia, the duration of hospitalization, NIHSS at the end of hospitalization and mRS at 3 months were compared between the negative parenchymal FLAIR and positive parenchymal FLAIR groups, and between study group and patients without 3 month follow-up, using the Chi² (χ^2) test for categorical variables, Student *t* test for normally distributed continuous variables and the Mann–Whitney U test for non-normally distributed continuous variables.

Interobserver agreement in determining parenchymal FLAIR positivity stroke lesions was estimated by Cohen's

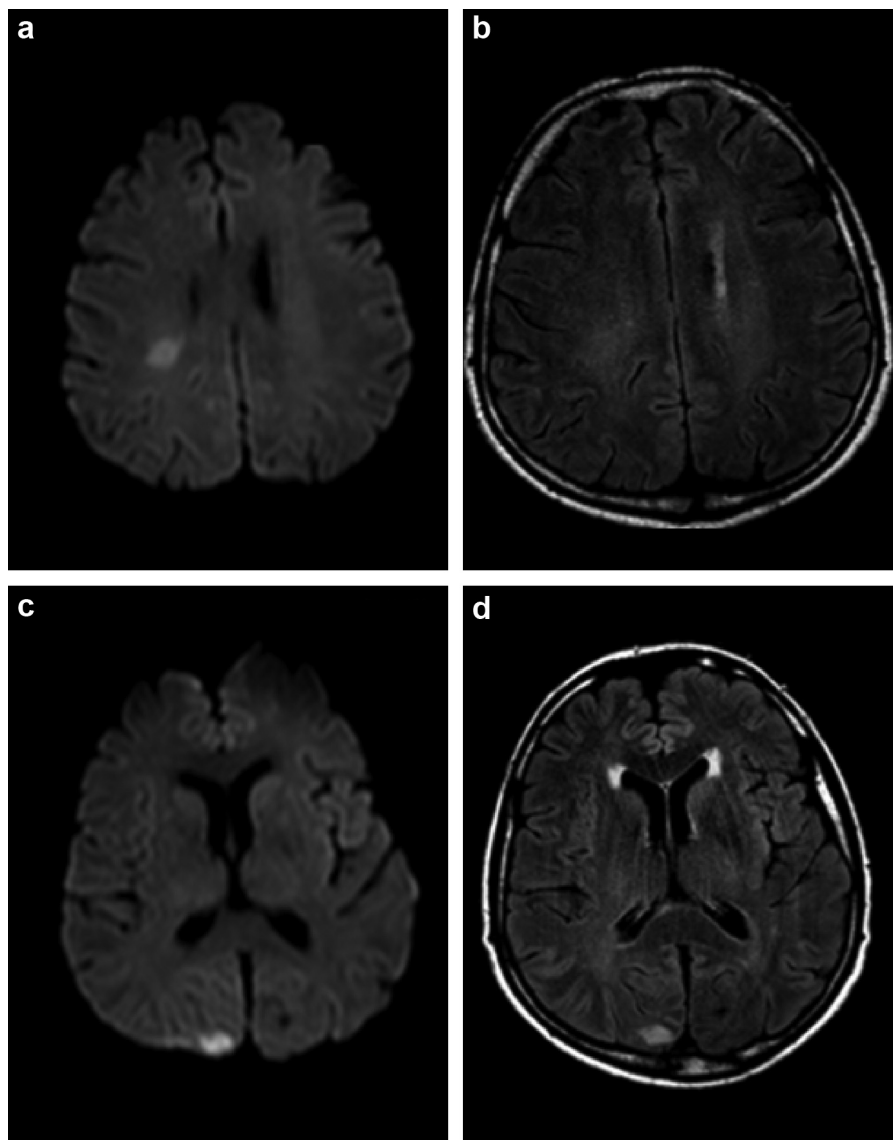


Figure 2 The patient presented two recent ischemic lesions on DWI, one in the right middle cerebral artery (a) and the second in the right posterior cerebral artery (c). But on FLAIR images, signals of those two lesions were different: a clear positive FLAIR signal for the posterior one (d) and a subtle FLAIR signal for the other (b). Regarding the difficulty to precisely date the two lesions and the posterior lesion, this patient was excluded.

Kappa coefficient. Logistic regression analysis was performed to identify independent predictors of NIHSS differences and for mRS at 3 months. The explanatory variables were identified by an univariate test: Mann-Whitney/Wilcoxon Test for NIHSS difference and Kruskal-Wallis Test for mRS. We finally estimated the predictive values (sensitivity, specificity, positive predictive value and negative predictive value) of the parenchymal FLAIR hyperintensity of stroke lesion to determine the good or poor outcomes at discharge and at 3 months.

The statistical analyses used Epi Info (version 3.5.3) and Matrix Laboratory (version 7.10.0) software. $P < 0.2$ was considered statistically significant for univariate test that determine predictors and $P < 0.05$ was considered statistically significant for other tests referring to conventional rules [21].

Results

Included patients

Between November 2007 and March 2012, 135 patients met the inclusion criteria. Fifteen of the 135 patients (11.1%) had major artifacts on FLAIR images and 34 of 135 patients (25.2%) did not have medical follow-up at 3 months. Two (1.5%) patients presented multiple lesions with different FLAIR signal. Finally, 51 (37.7%) patients were excluded. The study group comprised of 84 patients aged 18 to 91 years (median age: 64 years, [IQR (interquartile range): 50 to 75 years]), of which 48 were men (median age: 64 years, [IQR: 50 to 74 years]) and 36 were women (median age: 64 years, [IQR: 50 to 76 years]). Median time between stroke onset and imaging was 150 minutes, (IQR: 120 to

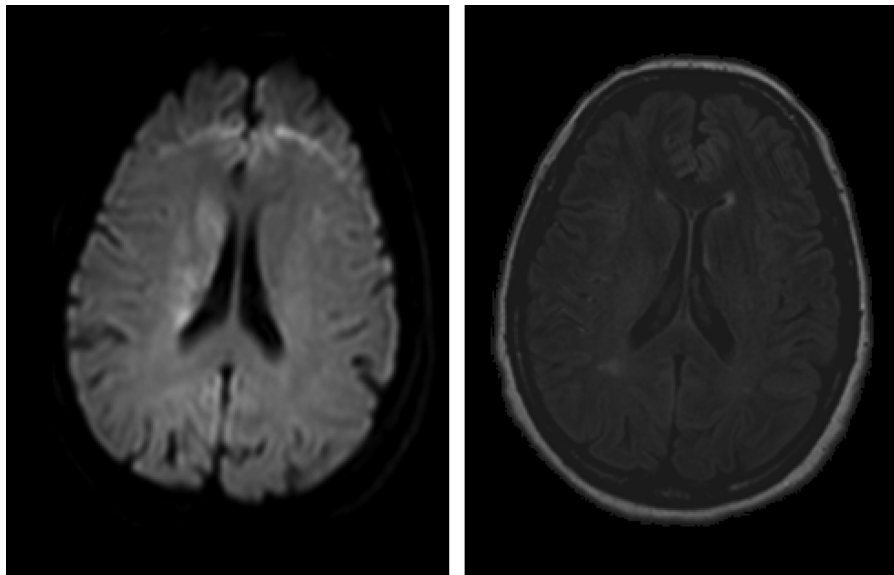


Figure 3 Patient with ischemic lesion in right middle cerebral territory, with no parenchymal FLAIR signal but with intravascular FLAIR hyperintensity.

Table 1 Group comparison of included patients and excluded patients because of the lack of medical follow-up for all clinical, biological and imaging characteristics.

	Included patients	Patients without medical follow-up	P-value
Age (years)	84	34	
ASPECT score	64 IQR [50–75]	70 IQR [64–74]	0.0439*
Stroke to MRI delay (mn)	—	—	0.169
Stroke to treatment delay (mn)	150 IQR [120–180]	172 [151–200]	0.0051*
Glycemia (g/L)	179 [150–205]	205 [180–215]	0.009*
Stroke subtype TOAST (n, %)	1.16 [1.03–1.35]	1.15 [0.74–1.3]	0.0374*
Stroke subtype TOAST (n, %)			0.2837
Large artery atherosclerosis	20 (23.8%)	10 (29.4%)	
Cardioembolism	24 (28.6%)	15 (44.1%)	
Small vessel occlusion	1 (1.2%)	0 (0%)	
Other determined etiology	12 (14.3%)	2 (5.9%)	
Undetermined etiology	27 (32.1%)	7 (20.6%)	
Fazekas' score			0.7745
0	38 (45.2%)	12 (35.3%)	
1	25 (29.8%)	11 (32.4%)	
2	13 (15.5%)	7 (20.6%)	
3	8 (9.5%)	12 (10.2%)	
Positive FLAIR (n, %)	38 (45.2%)	17 (50%)	0.394
NIHSS at admission	9.5 [4–14]	11 [6–17]	0.0254*
Gender (male: n, %)	48 (57.1%)	19 (55.9%)	0.53
Diastolic blood pressure (mmHg)	66 [47–72]	66.5 [45–74]	0.89
Systolic blood pressure (mmHg)	130 [109–140]	136 [109–150]	0.235
NIHSS at discharge	3 [0–8]	8 [5–15]	0.0004*
NIHSS difference (n, %)			0.2242
1	7 (8.3%)	0 (0%)	
2	1 (1.2%)	2 (5.9%)	
3	19 (22.6%)	12 (35.3%)	
4	24 (28.6%)	11 (32.4%)	
5	19 (22.6%)	5 (14.7%)	
6	10 (11.9%)	2 (5.9%)	
7	4 (4.8%)	2 (5.9%)	

* Significant values.

180 minutes) and median time to treatment was 179 minutes (IQR: 150 to 205 minutes).

Excluded patients

The 34 patients excluded from the study because of lack of medical follow-up were significantly younger, have larger delays between stroke onset and MRI and between stroke onset and treatment. They had lower glycemia and had higher NIHSS at admission and at discharge (Table 1).

Interobserver agreement

The interobserver agreement in the detection of parenchymal FLAIR hyperintensity was of 82% and $\kappa = 0.63$ (95% confidence interval [0.49–0.77]). A consensus was needed for 21 patients.

Group comparison

According to the readers' consensus, 46 of 84 patients (54.8%) had negative FLAIR and 38 of 84 (45.2%) had positive FLAIR results. According to the threshold chosen for univariate tests ($P < 0.2$), the 2 groups (FLAIR positive and FLAIR negative) were significantly different in terms of stroke-to-MRI delay ($P = 0.105$), stroke-to-treatment delay ($P = 0.025$), glycemia ($P = 0.05$) and 3 months mRS outcome ($P = 0.166$), but not NIHSS difference ($P = 0.53$) (Table 2).

Multivariate analysis for NIHSS

Logistic regression analysis performed to explore the predictors of NIHSS difference identified age, NIHSS at admission ($P = 0.042$) and systolic blood pressure ($P = 0.016$) as potential predictor variables but not parenchymal FLAIR visibility and duration of hospitalization. Multivariate analysis

Table 2 Group comparison of positive FLAIR and negative FLAIR for all clinical, biological and imaging characteristics.

	Positive FLAIR 38 (45.2%)	Negative FLAIR 46 (54.8%)	P-value
Age (years)	65.5 [49.0–76.0]	64 [53–74]	0.797
Gender (male, n, %)	22 (57.9%)	26 (56.5%)	0.538
NIHSS at admission	14 [10–20]	14 [9–19]	0.607
Hospitalisation duration (days)	12 [7–16]	8.5 [5–18]	0.457
Systolic blood pressure (mmHg)	140 [130–159]	140 [130–155]	0.97
Diastolic blood pressure (mmHg)	77 [69–80]	70 [66–84]	0.605
Glycemia (g/L)	1.22 [1.11–1.38]	1.12 [0.96–1.30]	0.05*
Stroke-to-MRI delay (mn)	168 [120–185]	143 [120–165]	0.105*
Stroke-to-treatment delay (mn)	190 [160–210]	162 [140–190]	0.025*
ASPECT score	8 [7–9]	8 [7.8]	0.682
Fazekas' score (n, %)			0.540
0	17 (44.7%)	21 (45.7%)	
1	12 (31.6%)	13 (28.3%)	
2	4 (10.5%)	9 (19.6%)	
3	5 (13.2%)	3 (6.5%)	
Stroke subtype TOAST (n, %)			0.747
Large artery atherosclerosis	11 (29.7%)	9 (20.0%)	
Cardioembolism	11 (29.7%)	13 (28.9%)	
Small vessel occlusion	0 (0%)	1 (2.2%)	
Other determined etiology	5 (13.5%)	7 (15.6%)	
Undetermined etiology	10 (27%)	15 (33.0%)	
NIHSS difference (n, %)			0.53
1	4 (10.5%)	3 (6.5%)	
2	1 (2.6%)	3 (6.5%)	
3	7 (18.4%)	15 (32.6%)	
4	14 (36.8%)	11 (23.9%)	
5	7 (18.4%)	6 (13%)	
6	5 (13.2%)	7 (15.2%)	
7	0 (0%)	1 (2.2%)	
mRS (n, %)			0.166*
0–1	5 (13.2%)	11 (23.9%)	
> 1	33 (86.8%)	35 (76.1%)	

* Significant values ($P < 0.2$).

Table 3 Logistic regression on NIHSS difference.

Group characteristics	Estimated odds ratio (95% CI)	P-values
Age	0.94 [0.87–1.02]	0.155
NIHSS at admission	0.80 [0.65–0.99]	0.042*
Parenchymal FLAIR positivity	0.59 [0.07–4.59]	0.615
Systolic blood pressure	0.91 [0.84–0.98]	0.016*

* Significant values ($P < 0.05$).

Table 4 Logistic regression on mRS.

Group characteristics	Estimated odds ratio (95% CI)	P-values
Age	0.92 [0.86–1.00]	0.054
Stroke-to-MRI delay	0.98 [0.95–1.01]	0.220
Stroke-to-treatment delay	0.99 [0.96–1.02]	0.606
Parenchymal FLAIR positivity	8.65 [1.20–62.0]	0.031*
Systolic blood pressure	1.00 [0.96–1.04]	0.824
NIHSS difference	2.75 [1.38–5.46]	0.003*

* Significant values ($P < 0.05$).

performed with the same variables and parenchymal FLAIR hyperintensity confirmed the findings above (Table 3).

Multivariate analysis for mRS

Parenchymal FLAIR hyperintensity, age, systolic blood pressure, stroke-to-MRI and stroke-to-treatment delays were identified as predictor variables for mRS at 3 months by univariate test. We also included NIHSS difference as a parameter to study its relation with mRS. The multivariate analysis on mRS determined that parenchymal FLAIR positivity (OR = 8.65, 95% CI: [1.20–62.0], $P = 0.031$) and NIHSS difference (OR = 2.75 95% CI: [1.38–5.46], $P = 0.003$) were significant independent predictors of mRS (Table 4).

Predictive values

Parenchymal FLAIR hyperintensity of the ischemic lesion identified poor outcomes at 3 months with a Se of 0.49 (95% CI: [0.37–0.60]), a Sp of 0.69 (95% CI: [0.46–0.91]), a PPV of 0.87 (95% CI: [0.76–0.98]) and a NPV of 0.24 (95% CI: [0.12–0.36]).

Discussion

The main finding of the present study is that the parenchymal FLAIR hyperintensity of ischemic stroke before thrombolysis has a prognostic value. It is not a predictive factor of early outcome but its presence is predictive of a worst outcome at 3 months. Our findings are similar to

previously published data on this predictive association of parenchymal FLAIR hyperintensity and 3 months outcomes but we highlight the importance of the parenchymal signal alone (i.e., not associated to intravascular signal) [7–9,22]. Further, we have also established that parenchymal FLAIR hyperintensity of stroke lesion is not only unipredictive of NIHSS at day 1 of thrombolysis, but also unipredictive of NIHSS at the end of hospitalization corresponding to the end of the acute phase of stroke which could be clinically instable phase [9].

Ebinger et al. demonstrated that FLAIR hyperintensities in arteries and parenchyma are associated with poor outcome [8]. It is important to note that parenchymal and vascular FLAIR hyperintensities correspond to two very different processes [23]. While the parenchymal FLAIR hyperintensity corresponds to parenchymal lesion, vascular FLAIR hyperintensity corresponds to a slow blood flow in connection with collateral circulation [7,14]. In the present study, our analysis was limited to parenchymal FLAIR hyperintensities only. Our data clearly demonstrates the prognostic value of parenchymal FLAIR hyperintensities alone.

The results of this study show that parenchymal FLAIR hyperintensity can predict poor outcome with a high PPV. In 87% of cases, when parenchymal FLAIR hyperintensity was observed, the outcome at 3 months was poor, even if they were treated by thrombolysis. We also demonstrated that the parenchymal FLAIR hyperintensity has moderate sensitivity, moderate specificity and low NPV. A large proportion of patients (51%) who had a poor outcome at 3 months presented negative parenchymal FLAIR. This suggests that although parenchymal FLAIR hyperintensity can predict 3 months outcomes, it may not be sufficient alone. Moreover, in a previous study, it was demonstrated that a negative parenchymal FLAIR can determine lesion age of < 4.5 hours with a high PPV and low NPV. This study shows that a positive parenchymal FLAIR can be used to predict poor outcome at 3 months with high PPV and low NPV which means that FLAIR hyperintensity also have a prognostic value [1].

Additionally, we also found that parenchymal FLAIR was not predictive of NIHSS difference, but systolic blood pressure and NIHSS at admission were. A previous study determined that high systolic blood pressure level during the hyperacute stage of ischemic stroke was associated with increased neurological impairment and poor functional outcome. Our data are confirmatory of these findings. There also was no difference noted in terms of length of hospitalization between NIHSS difference groups, which could potentially be a confusing factor. However, it is possible that patients with longer hospitalization probably had more spontaneous recovery and physiotherapy session explaining the lack of difference noted above. Thus, parenchymal FLAIR positivity is neither predictive of NIHSS at day 1 after thrombolysis, nor at the end of hospitalization [9].

The independent predictive variables of 3 months mRS were parenchymal FLAIR hyperintensity and NIHSS difference both of which were strongly correlated to poor outcome. Although there was no correlation between parenchymal FLAIR and NIHSS difference, they both were predictive of mRS at 3 months. A previous study highlighted the predictive value of NIHSS for mRS, just as hyperglycemia

for patients treated by thrombectomy [22]. In our study, hyperglycemia did not emerge as a predictive factor.

The links between parenchymal FLAIR hyperintensity, NIHSS difference and 3 months mRS is obvious. An ischemic lesion could have both irreversibly damaged areas demonstrating parenchymal FLAIR hyperintensity and some hypoperfused regions not visible on diffusion or FLAIR sequences [13]. While the irreversibly damaged lesions explain the poor mRS outcomes at 3 months, the ability to reperfuse the hypoperfused areas by IV or IA treatment can modify the NIHSS scores. Furthermore, subtle neurological recoveries can modify NIHSS scale while mRS is not modified [16]. Hence, NIHSS difference is not predictable by parenchymal FLAIR hyperintensity whereas mRS at 3 months is.

Our study has several limitations: the retrospective nature of our analysis being the first. Further prospective studies will be required to conclusively establish the prognostic value of parenchymal FLAIR hyperintensity of stroke. Further, the relatively small sample size could also increase the chances of a type II error. The second limitation is that the visual classification of parenchymal FLAIR lesions was based on dichotomization between positive and negative groups, which could be difficult to assess for some patients resulting in a substantial interobserver agreement. However, our interobserver agreement is similar to previous study based on FLAIR positivity [17]. This difficulty was limited by obtaining a consensus between the two readers. Further studies can be proposed to evaluate the benefit of signal quantification. The third limitation is the non-blinded reading which can lead to some bias. However, we made this methodological choice to be closer to the clinical practice. Another limitation of our study is that we focused on middle cerebral artery stroke only, and our results cannot be extrapolated to the anterior cerebral artery and vertebrobasilar territories. Further studies are needed for these territories. Finally, we may have an under-representativity of favorable mRS. It could possibly be explained by the fact that those patients could be less inclined to their medical follow-up by the neurologist. But we demonstrated that patients without follow-up were older and with higher neurological impairment suggesting that they could be followed in reeducation centers.

Our present study improves the understanding of parenchymal FLAIR hyperintensity as a prognostic factor of stroke outcome after thrombolysis. At 3 Tesla, the presence of hyperintense parenchymal FLAIR signal is predictive of a poor clinical outcome at 3 months. Our results raise the question, should parenchymal FLAIR hyperintense stroke lesions be thrombolysed? Is there a higher risk of hemorrhages? Our data suggests that if parenchymal FLAIR is positive, poor outcomes can be predicted but in case of negative parenchymal FLAIR, the outcome is not predictable. Parenchymal FLAIR associated with other imaging, clinical and biological prognostic factors could help to select more finely eligible patient to thrombolysis.

Conclusion

To date, the usefulness of parenchymal FLAIR images helped to predict a stroke onset less than 4.5 hours when

negative, thereby allowing to date wake-up strokes. Our study demonstrated the prognostic value of this imaging sequence suggesting that the thrombolysis of hyperintense parenchymal FLAIR of acute stroke doesn't improve functional prognosis.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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